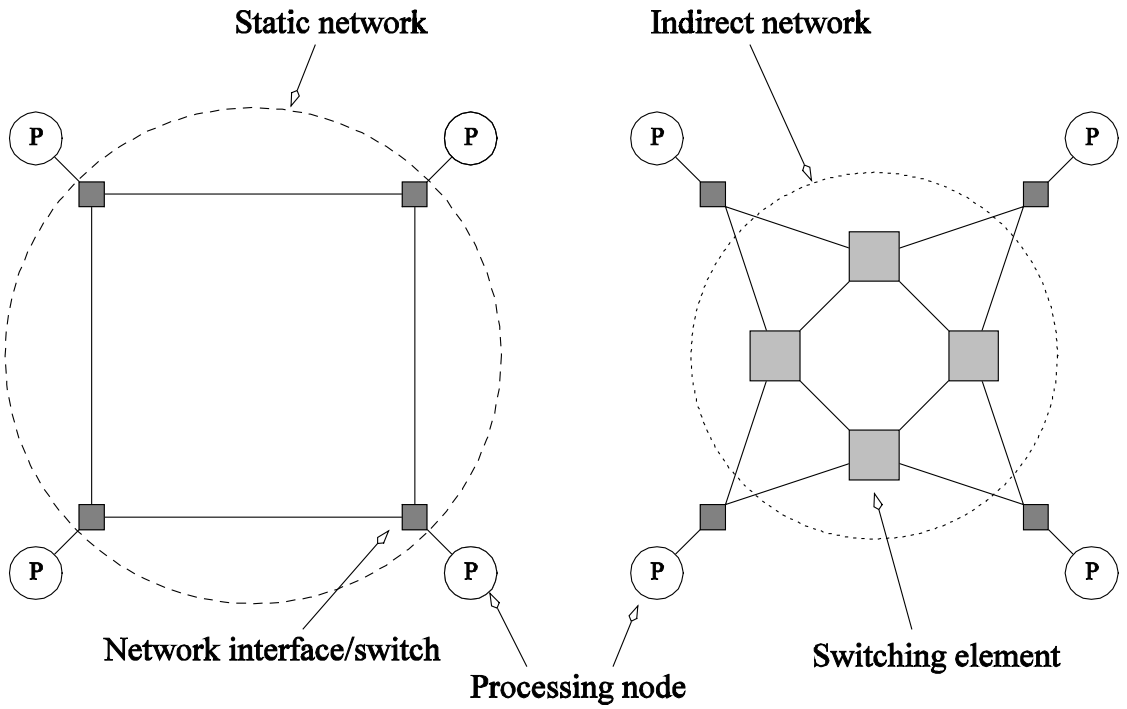
Interconnection Networks for Parallel Computers

- Interconnection networks carry data between processors and to memory.
- Interconnects are made of switches and links (wires, fiber).
- Interconnects are classified as static or dynamic.
- Static networks consist of point-to-point communication links among processing nodes and are also referred to as *direct* networks.
- Dynamic networks are built using switches and communication links. Dynamic networks are also referred to as *indirect* networks.

Static and Dynamic Interconnection Networks



Classification of interconnection networks: (a) a static network; and (b) a dynamic network.

Interconnection Networks

- Switches map a fixed number of inputs to outputs.
- The total number of ports on a switch is the *degree* of the switch.
- The cost of a switch grows as the square of the degree of the switch, the peripheral hardware linearly as the degree, and the packaging costs linearly as the number of pins.

Interconnection Networks: Network Interfaces

- Processors talk to the network via a network interface.
- The network interface may hang off the I/O bus or the memory bus.
- In a physical sense, this distinguishes a cluster from a tightly coupled multicomputer.
- The relative speeds of the I/O and memory buses impact the performance of the network.

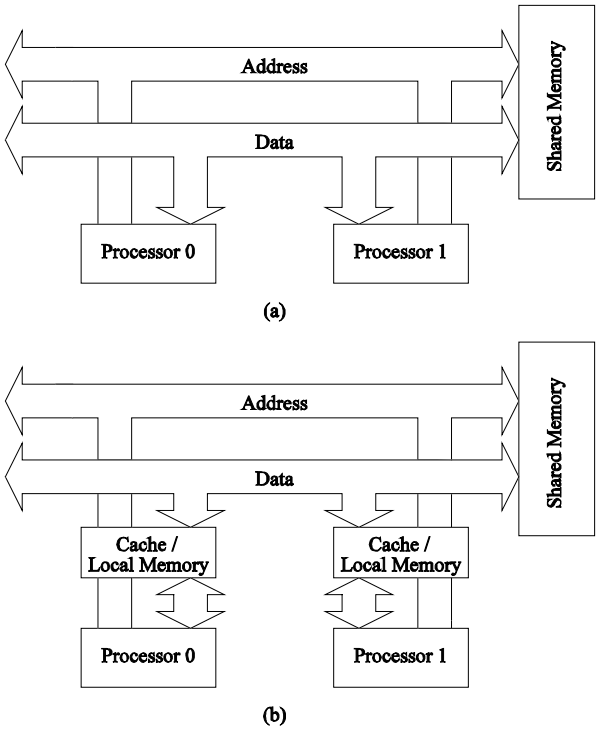
Network Topologies

- A variety of network topologies have been proposed and implemented.
- These topologies tradeoff performance for cost.
- Commercial machines often implement hybrids of multiple topologies for reasons of packaging, cost, and available components.

Network Topologies: Buses

- Some of the simplest and earliest parallel machines used buses.
- All processors access a common bus for exchanging data.
- The distance between any two nodes is *O(1)* in a bus. The bus also provides a convenient broadcast media.
- However, the bandwidth of the shared bus is a major bottleneck.
- Typical bus based machines are limited to dozens of nodes. Sun Enterprise servers and Intel Pentium based shared-bus multiprocessors are examples of such architectures.

Network Topologies: Buses

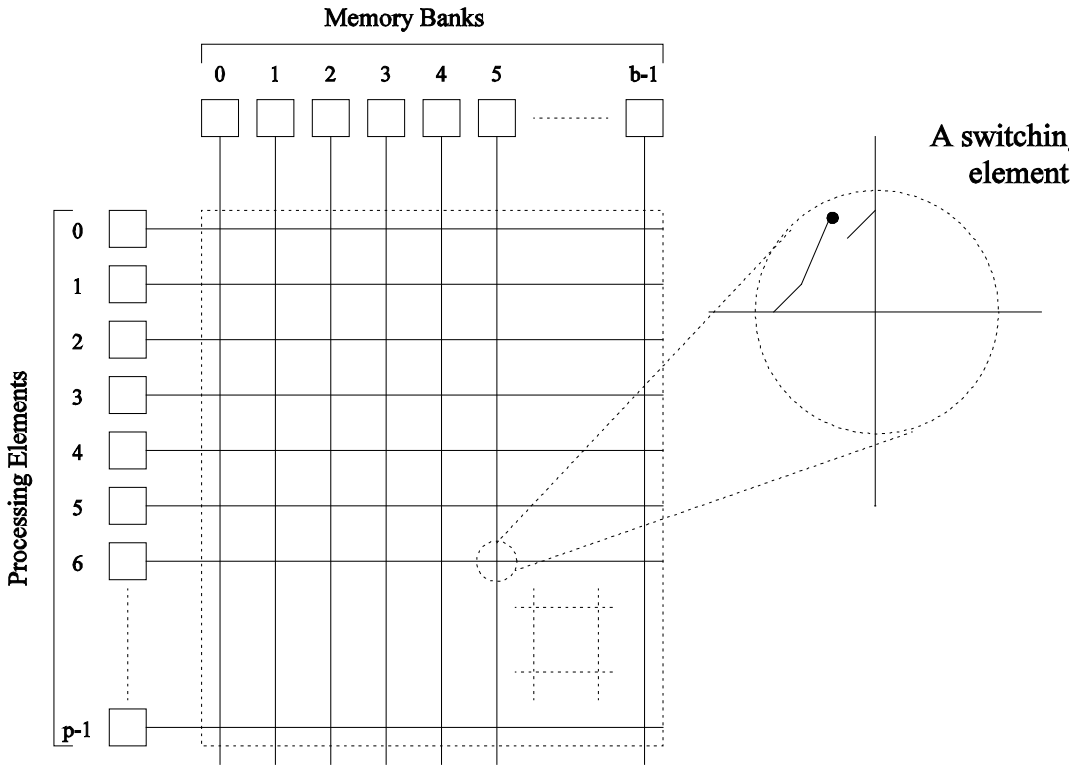


Bus-based interconnects (a) with no local caches; (b) with local memory/caches.

Since much of the data accessed by processors is local to the processor, a local memory can improve the performance of bus-based machines.

Network Topologies: Crossbars

A crossbar network uses an $p \times m$ grid of switches to connect p inputs to m outputs in a non-blocking manner.



A completely non-blocking crossbar network connecting p processors to b memory banks.

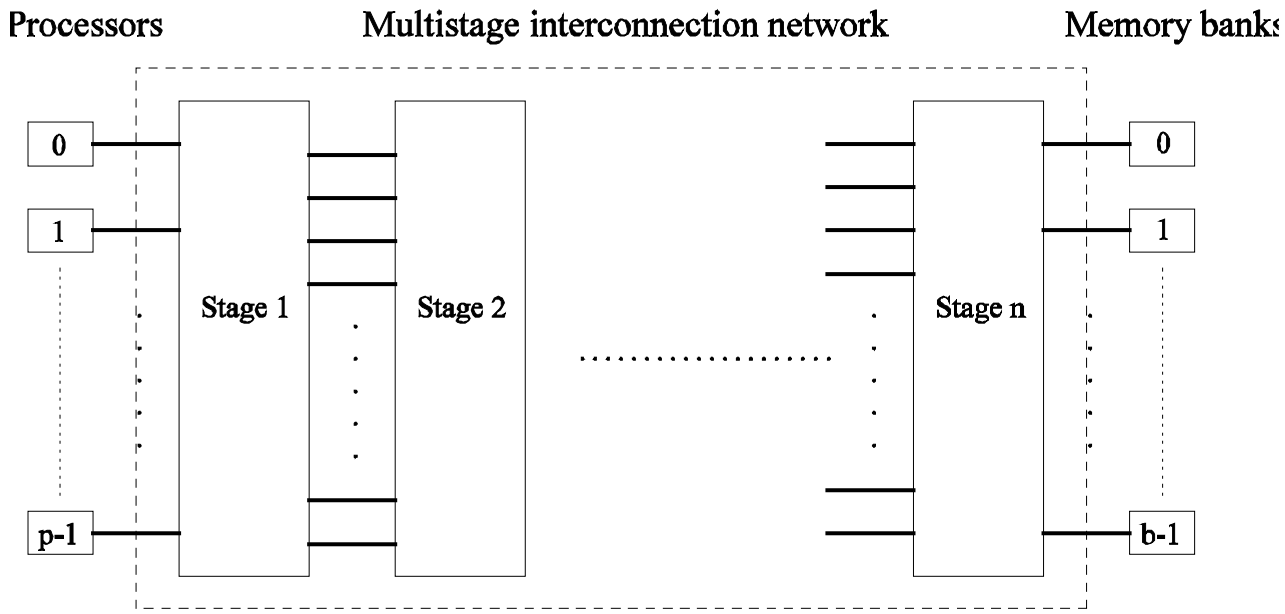
Network Topologies: Crossbars

- The cost of a crossbar of p processors grows as $O(p^2)$.
- This is generally difficult to scale for large values of p.
- Examples of machines that employ crossbars include the Sun Ultra HPC 10000 and the Fujitsu VPP500.

Network Topologies: Multistage Networks

- Crossbars have excellent performance scalability but poor cost scalability.
- Buses have excellent cost scalability, but poor performance scalability.
- Multistage interconnects strike a compromise between these extremes.

Network Topologies: Multistage Networks



The schematic of a typical multistage interconnection network.

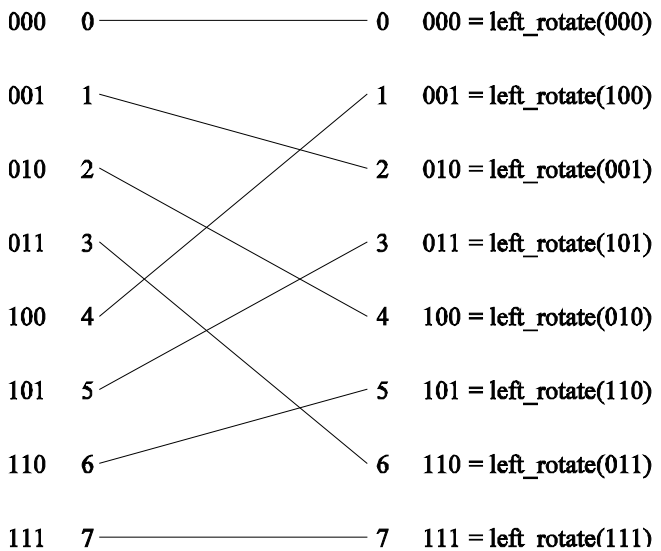
Network Topologies: Multistage Omega Network

- One of the most commonly used multistage interconnects is the Omega network.
- This network consists of *log p* stages, where *p* is the number of inputs/outputs.
- At each stage, input *i* is connected to output *j* if:

$$j = \begin{cases} 2i, & 0 \leq i \leq p/2 - 1 \\ 2i + 1 - p, & p/2 \leq i \leq p - 1 \end{cases}$$

Network Topologies: Multistage Omega Network

Each stage of the Omega network implements a perfect shuffle as follows:



A perfect shuffle interconnection for eight inputs and outputs.

Network Topologies: Multistage Omega Network

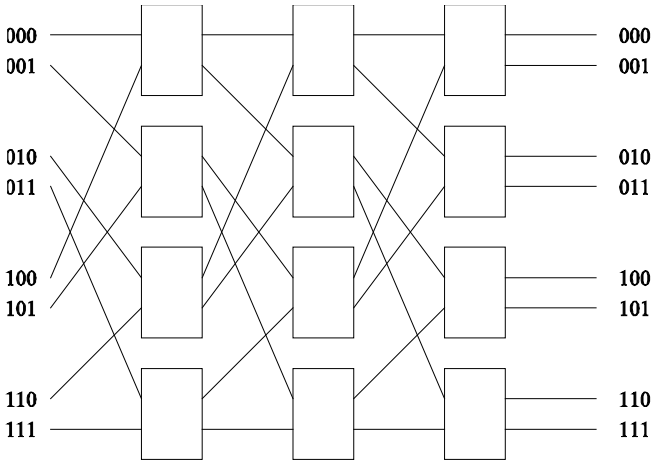
- The perfect shuffle patterns are connected using 2×2 switches.
- The switches operate in two modes – crossover or passthrough.



Two switching configurations of the 2 × 2 switch:
(a) Pass-through; (b) Cross-over.

Network Topologies: Multistage Omega Network

A complete Omega network with the perfect shuffle interconnects and switches can now be illustrated:



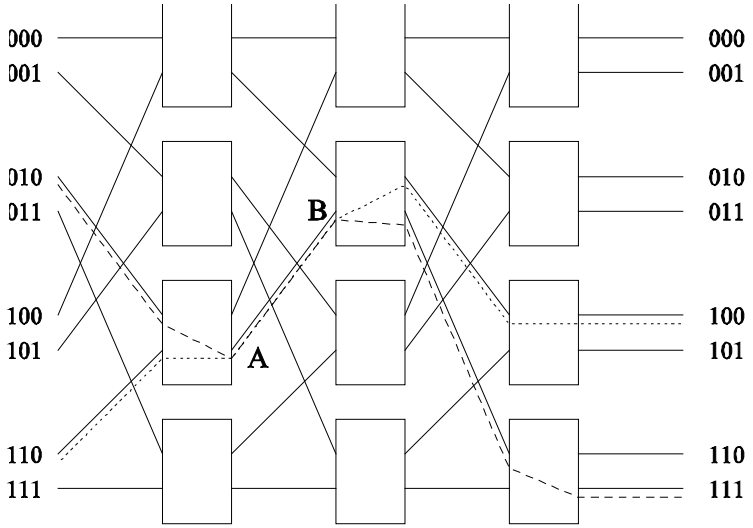
A complete omega network connecting eight inputs and eight outputs.

An omega network has $p/2 \times \log p$ switching nodes, and the cost of such a network grows as $(p \log p)$.

Network Topologies: Multistage Omega Network – Routing

- Let *s* be the binary representation of the source and *d* be that of the destination processor.
- The data traverses the link to the first switching node. If the most significant bits of *s* and *d* are the same, then the data is routed in pass-through mode by the switch else, it switches to crossover.
- This process is repeated for each of the *log p* switching stages.
- Note that this is not a non-blocking switch.

Network Topologies: Multistage Omega Network – Routing



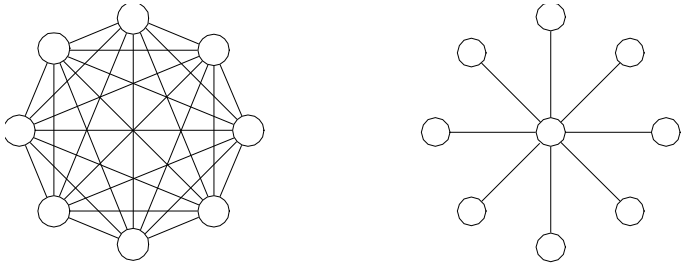
An example of blocking in omega network: one of the messages (010 to 111 or 110 to 100) is blocked at link AB.

Network Topologies: Completely Connected Network

- Each processor is connected to every other processor.
- The number of links in the network scales as $O(p^2)$.
- While the performance scales very well, the hardware complexity is not realizable for large values of p.
- In this sense, these networks are static counterparts of crossbars.

Network Topologies: Completely Connected and Star Connected Networks

Example of an 8-node completely connected network.



(a) A completely-connected network of eight nodes;
(b) a star connected network of nine nodes.

Network Topologies: Star Connected Network

- Every node is connected only to a common node at the center.
- Distance between any pair of nodes is *O(1).* However, the central node becomes a bottleneck.
- In this sense, star connected networks are static counterparts of buses.